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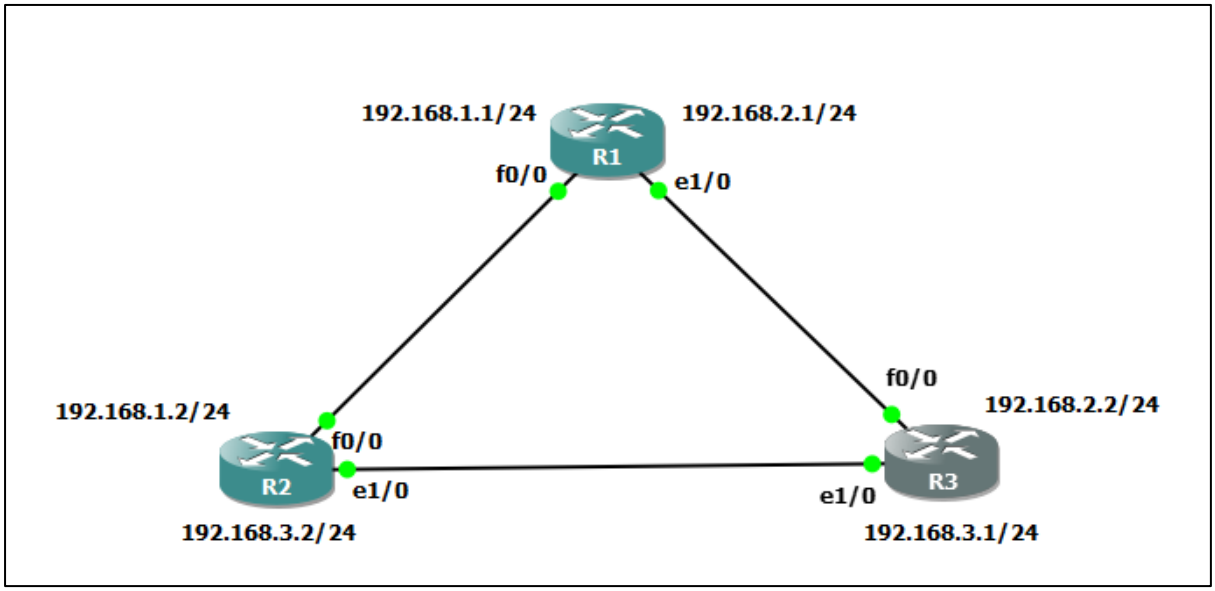
Practical No 1:

Aim: Configure IP SLA tracking and path control topology



Step 1: Config all routers with their respective IP address and Loopback address according to the topology.

R1

```
R1#
R1#conf t
Enter configuration commands, one per line.  End with
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
```

R1

```
*Apr 19 19:02:01.303: %LINEPROTO-5-UPDOWN: Line protocol c
et0/0, changed state to up
R1(config)#int e1/0
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shut
```

R2

```
R2#
R2#conf t
Enter configuration commands, one per line.  End with
R2(config)#int f0/0
R2(config-if)#ip address 192.168.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
```

```
R2
et0/0, changed state to up
R2(config)#int e1/0
R2(config-if)#ip address 192.168.3.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
```

```
R3

R3(config)#int f0/0
R3(config-if)#ip address 192.168.2.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#e
*Apr 19 19:06:04.339: %LINK-3-UPDOWN: Interface FastEth
o up
R3(config-if)#exit
R3(config)#
*Apr 19 19:06:04.339: %ENTITY_ALARM-6-INFO: CLEAR INFO
istrative State Down
*Apr 19 19:06:05.419: %LINEPROTO-5-UPDOWN: Line protoco
et0/0, changed state to up
R3(config)#int e1/0
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
```

Step 2: To connect with R2 N/W

```
R2
0, changed state to up
R2(config)#ip route 192.168.2.0 255.255.255.0 192.168.1.1
R2(config)#ip route 192.168.2.0 255.255.255.0 192.168.3.1
R2(config)#track 10 ip route 192.168.2.0 255.255.255.0 reachability
R2(config-track)#exit
R2(config)#exit
R2#
```

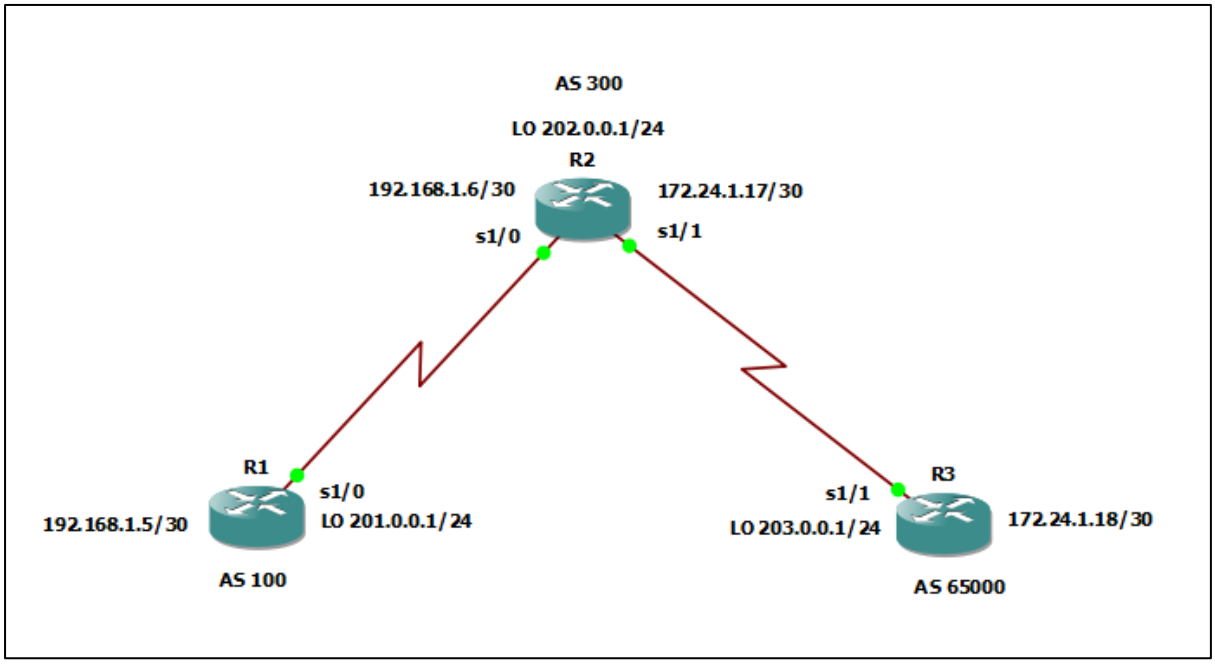
Step 3: Verify route connection

R2

```
R2#
*Apr 19 19:07:47.923: %SYS-5-CONFIG_I: Configured from console by console
R2#show track
Track 10
  IP route 192.168.2.0 255.255.255.0 reachability
  Reachability is Up (static)
    1 change, last change 00:00:12
  First-hop interface is FastEthernet0/0
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 192.168.3.1
                    [1/0] via 192.168.1.1
C    192.168.3.0/24 is directly connected, Ethernet1/0
R2#
```

Practical No 2:

Aim: Using the AS_path attribute



Step 1: Config all routers with their respective IP address and Loopback address according to the topology.

```
R1
*Apr 19 11:05:40.147: %LINEPROTO-5-UPDOWN: Line protocol on In
 changed state to down
*Apr 19 11:05:40.147: %LINEPROTO-5-UPDOWN: Line protocol on In
 changed state to down
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#INT S1/0
R1(config-if)#NO IP ADDRESS
R1(config-if)#IP ADDRESS 192.168.1.5 255.255.255.252
R1(config-if)#NO SHUT
R1(config-if)#EXIT
R1(config)#INT LO0
R1(config-if)#IP ADDRESS 201.0.0.1 255.255.255.0
R1(config-if)#NO SHUT
R1(config-if)#EXIT
R1(config)#
```

R2

```
*Apr 19 11:05:40.143: %LINEPROTO-5-UPDOWN: Line protocol on In
 changed state to down
*Apr 19 11:05:40.147: %LINEPROTO-5-UPDOWN: Line protocol on In
 changed state to down
R2#
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#INT S1/0
R2(config-if)#IP ADDRESS 192.168.1.6 255.255.255.252
R2(config-if)#NO SHUT
R2(config-if)#EXIT
R2(config)#INT S1/1
R2(config-if)#IP ADDRESS 172.24.1.17 255.255.255.252
R2(config-if)#NO SHUT
R2(config-if)#EXIT
R2(config)#INT LO0
R2(config-if)#IP ADDRESS 202.0.0.1 255.255.255.0
R2(config-if)#NO SHUT
R2(config-if)#EXIT
R2(config)#
```

R3

```
R3#
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#INT S1/1
R3(config-if)#IP ADDRESS 172.24.1.18 255.255.255.252
R3(config-if)#NO SHUT
R3(config-if)#EXIT
R3(config)#INT LO0
R3(config-if)#IP ADDRESS 203.0.0.1 255.255.255.0
R3(config-if)#NO SHUT
R3(config-if)#EXIT
```

Step 2: Provide BGP connection to all routers according to their AS Path and neighbors IP address and network loopback address.

R1

```
 changed state to up
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router bgp 100
R1(config-router)#no sync
R1(config-router)#neighbor 192.168.1.6 remote-as 300
R1(config-router)#network 201.0.0.0
R1(config-router)#exit
R1(config)#
```

```
R2
 changed state to up
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router bgp 300
R2(config-router)#no sync
R2(config-router)#neighbor 192.168.1.5 remote-as 100
R2(config-router)#neighbor 172.24.1.18 remote-as 1
*Apr 19 11:13:11.407: %BGP-5-ADJCHANGE: neighbor 192.168.1.5 Up
R2(config-router)#neighbor 172.24.1.18 remote-as 65000
R2(config-router)#network 202.0.0.0
R2(config-router)#exit
R2(config)#
```

```
R3
*Apr 19 11:07:53.215: %SYS-5-CONFIG_I: Configured from console by
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router bgp 65000
R3(config-router)#no sync
R3(config-router)#neighbor 172.24.1.17 remote-as 300
R3(config-router)#network 203.0.0.0
R3(config-router)#
*Apr 19 11:15:21.119: %BGP-5-ADJCHANGE: neighbor 172.24.1.17 Up
R3(config-router)#exit
R3(config)#exit
```

Step 3: Run command “show ip bgp neighbors” to see the neighbours connection and “show ip route” or “show ip bgp” to see the network configured using BGP.

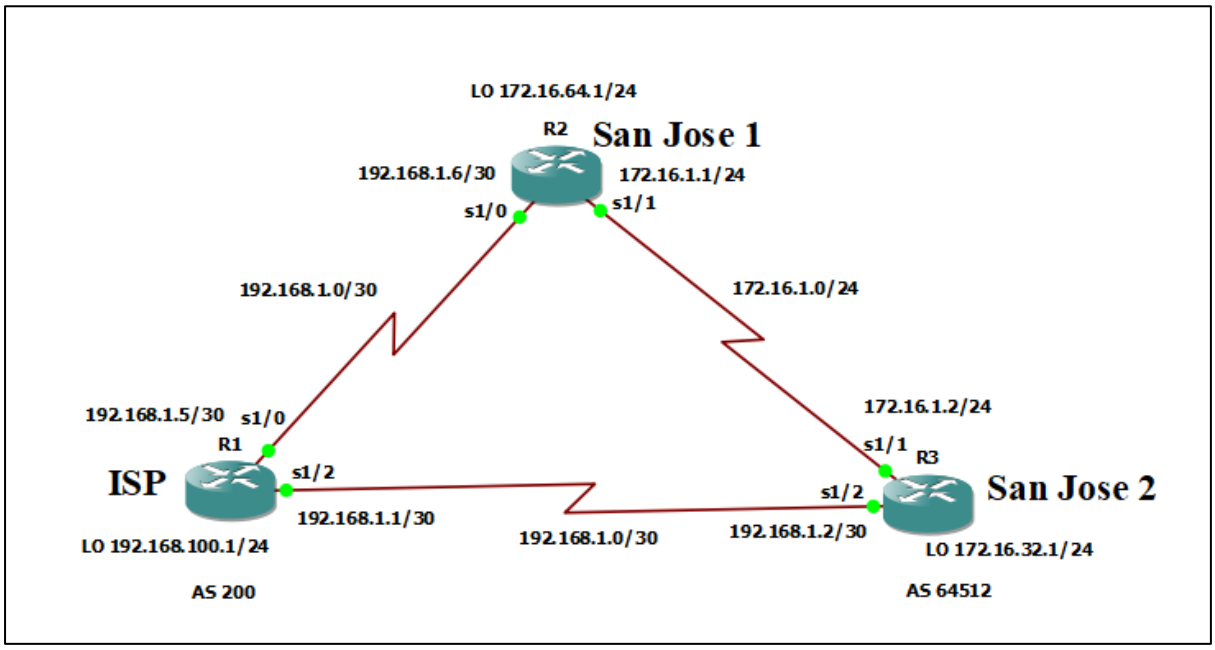
```
R1
R1#sho
*Apr 19 11:16:38.463: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip bgp neighbors
BGP neighbor is 192.168.1.6,  remote AS 300, external link
  BGP version 4, remote router ID 202.0.0.1
  BGP state = Established, up for 00:03:29
  Last read 00:00:28, last write 00:00:28, hold time is 180, keepalive interval
is 60 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(old & new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0
                         Sent       Rcvd
    Opens:                  1          1
    Notifications:          0          0
    Updates:                1          3
    Keepalives:             5          5
    Route Refresh:          0          0
    Total:                  7          9
  Default minimum time between advertisement runs is 30 seconds

 For address family: IPv4 Unicast
```

```
R1
Datagrams (max data segment is 1460 bytes):
Rcvd: 11 (out of order: 0), with data: 7, total data bytes: 286
Sent: 14 (retransmit: 0, fastretransmit: 0, partialack: 0, Second Congestion: 0)
, with data: 7, total data bytes: 192
R1#
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    201.0.0.0/24 is directly connected, Loopback0
B    202.0.0.0/24 [20/0] via 192.168.1.6, 00:03:34
B    203.0.0.0/24 [20/0] via 192.168.1.6, 00:01:20
     192.168.1.0/30 is subnetted, 1 subnets
C       192.168.1.4 is directly connected, Serial1/0
```

```
R1
.....
Success rate is 0 percent (0/5)
R1#show ip bgp
BGP table version is 4, local router ID is 201.0.0.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 201.0.0.0        0.0.0.0                  0         32768 i
*> 202.0.0.0        192.168.1.6              0             0 300 i
*> 203.0.0.0        192.168.1.6                            0 300 65000 i
R1#
```

Practical No 3:

Aim: Configuring IBGP and EPGP sessions, local preferences, and MED.



Step 1: Config all routers with their respective IP address and Loopback address according to the topology and also apply hostname and timeout commands accordingly clock rates.

R1

```
 changed state to down
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname ISP
ISP(config)#no ip domain-lookup
ISP(config)#line con 0
ISP(config-line)#logging synchronous
ISP(config-line)#exec-timeout 0 0
ISP(config-line)#exit
ISP(config)#interface Loopback0
ISP(config-if)#ip address 192.168.100.1 255.255.255.0
ISP(config-if)#exit
ISP(config)#interface Serial1/0
ISP(config-if)#ip address 192.168.1.5 255.255.255.252
ISP(config-if)#clockrate 128000
ISP(config-if)#no shutdown
ISP(config-if)#exit
ISP(config)#interface Serial1/2
ISP(config-if)#ip address 192.168.1.1 255.255.255.252
ISP(config-if)#no shutdown
ISP(config-if)#end
ISP#exit
```

R2

```
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname SanJose1
SanJose1(config)#no ip domain-lookup
SanJose1(config)#line con 0
SanJose1(config-line)#logging synchronous
SanJose1(config-line)#exec-timeout 0 0
SanJose1(config-line)#exit
SanJose1(config)#interface Loopback0
SanJose1(config-if)#ip address 172.16.64.1 255.255.255.0
SanJose1(config-if)#exit
SanJose1(config)#interface Serial1/0
SanJose1(config-if)#ip address 192.168.1.6 255.255.255.252
SanJose1(config-if)#no shutdown
SanJose1(config-if)#exit
SanJose1(config)#interface Serial1/1
SanJose1(config-if)#ip address 172.16.1.1 255.255.255.0
SanJose1(config-if)#clockrate 128000
SanJose1(config-if)#no shutdown
SanJose1(config-if)#end
```

R3

```
R3#
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#hostname SanJose2
SanJose2(config)#no ip domain-lookup
SanJose2(config)#line con 0
SanJose2(config-line)#logging synchronous
SanJose2(config-line)#exec-timeout 0 0
SanJose2(config-line)#exit
SanJose2(config)#interface Loopback0
SanJose2(config-if)#ip address 172.16.32.1 255.255.255.0
SanJose2(config-if)#exit
SanJose2(config)#interface Serial1/2
SanJose2(config-if)#ip address 192.168.1.2 255.255.255.252
SanJose2(config-if)#clockrate 128000
SanJose2(config-if)#no shutdown
SanJose2(config-if)#exit
SanJose2(config)#interface Serial1/1
SanJose2(config-if)#ip address 172.16.1.2 255.255.255.0
SanJose2(config-if)#no shutdown
SanJose2(config-if)#end
```

Step 2: Configure EIGRP between the SanJose1 and SanJose2 routers.

R2

```
SanJose1#conf t
Enter configuration commands, one per line.
SanJose1(config)#router eigrp 1
SanJose1(config-router)#network 172.16.0.0
SanJose1(config-router)#
```

R3

```
 changed state to up
SanJose2#conf t
Enter configuration commands, one per line.
SanJose2(config)#router eigrp 1
SanJose2(config-router)#network 172.16.0.0
SanJose2(config-router)#
```

Step 3: Configure IBGP and verify BGP neighbours.

R2

```
ia11/1) is up: new adjacency
SanJose1(config-router)#exit
SanJose1(config)#router bgp 64512
SanJose1(config-router)#neighbor 172.16.32.1 remote-as 64512
SanJose1(config-router)#neighbor 172.16.32.1 update-source lOO
SanJose1(config-router)#
```

R3

```
SanJose2(config-router)#exit
SanJose2(config)#router bgp 64512
SanJose2(config-router)#neighbor 172.16.64.1 remote-as 64512
SanJose2(config-router)#neighbor 172.16.64.1
*Apr 19 12:31:43.507: %BGP-5-ADJCHANGE: neighbor 172.16.64.1 Up
SanJose2(config-router)#neighbor 172.16.64.1 update-source LOO
SanJose2(config-router)#exit
```

Step 4: Configure EBGP and verify BGP neighbors.

R1

```
ISP#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP(config)#router bgp 200
ISP(config-router)#neigh 192.168.1.6 remote-as 64512
ISP(config-router)#neigh 192.168.1.2 remote-as 64512
ISP(config-router)#network 192.168.100.0
ISP(config-router)#
*Apr 19 12:40:46.567: %BGP-5-ADJCHANGE: neighbor 192.168.1.6 Up
ISP(config-router)#
*Apr 19 12:41:47.931: %BGP-5-ADJCHANGE: neighbor 192.168.1.2 Up
ISP(config-router)#exit
ISP(config)#exit
```

R2

```
SanJose1(config-router)#
*Apr 19 12:31:43.635: %BGP-5-ADJCHANGE: neighbor 172.16.3
SanJose1(config-router)#exit
SanJose1(config)#ip route 172.16.0.0 255.255.0.0 null0
SanJose1(config)#router bgp 64512
SanJose1(config-router)#neigh 192.168.1.5 remote-as 200
SanJose1(config-router)#network 172.16.0.0
SanJose1(config-router)#
*Apr 19 12:41:26.671: %BGP-5-ADJCHANGE: neighbor 192.168.
SanJose1(config-router)#
```

R3

```
SanJose2(config-router)#neighbor 172.16.64.1 update-source LO0
SanJose2(config-router)#exit
SanJose2(config)#ip route 172.16.0.0 255.255.0.0 null0
SanJose2(config)#router bgp 64512
SanJose2(config-router)#neigh 192.168.1.1 remote-as 200
SanJose2(config-router)#network 172.16.0.
*Apr 19 12:42:27.879: %BGP-5-ADJCHANGE: neighbor 192.168.1.1 Up
SanJose2(config-router)#network 172.16.0.0
SanJose2(config-router)#
SanJose2(config-router)#exit
```

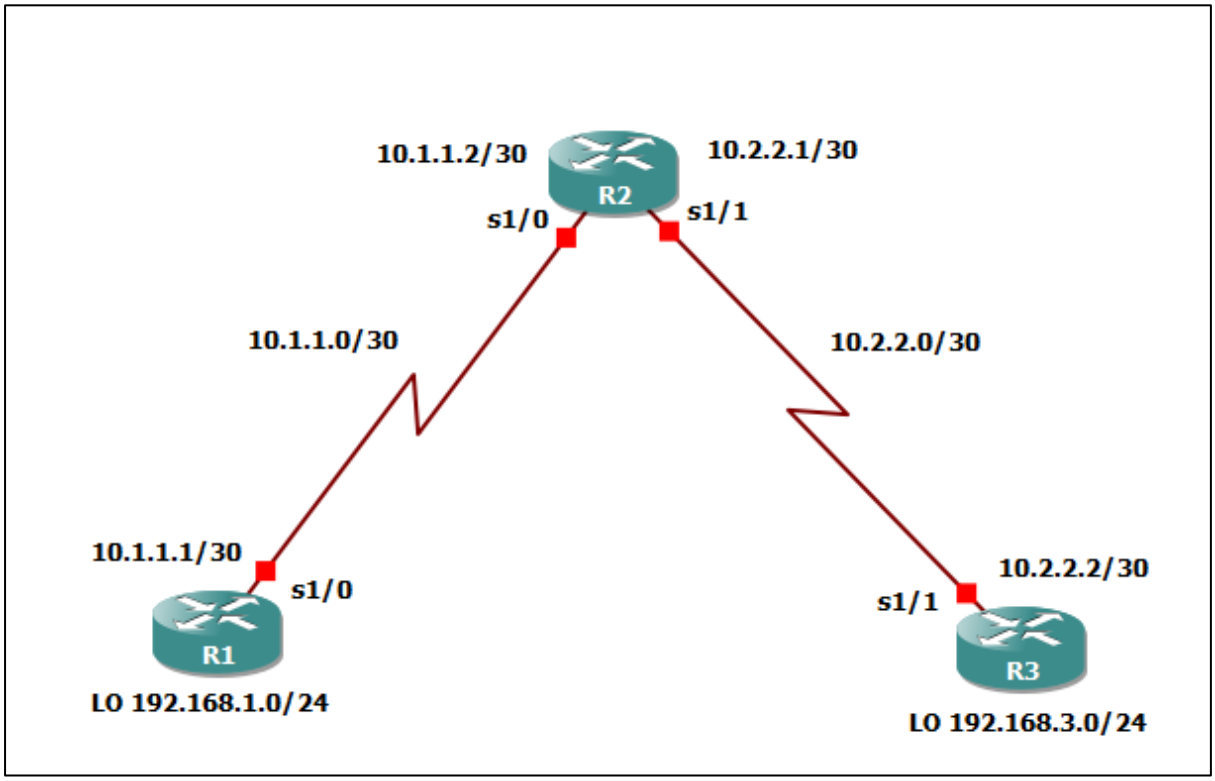
Step 5: Ping and verify from ISP to San Jose i.e EBGP to IBGP:

R1

```
ISP#
*Apr 19 12:43:00.563: %SYS-5-CONFIG_I: Configured from co
ISP#ping 172.16.1.1 source 192.168.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.1, timeout is 
Packet sent with a source address of 192.168.100.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max
ISP#ping 172.16.32.1 source 192.168.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.32.1, timeout is
Packet sent with a source address of 192.168.100.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max
ISP#ping 172.16.1.2 source 192.168.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 
Packet sent with a source address of 192.168.100.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max
ISP#
```

Practical No 4:

Aim: Secure the Management Plane.



Step 1: Configure loopbacks and assign addresses.

```
R1
R1#conf t
Enter configuration commands, one per line.  End with CNT
R1(config)#interface Loopback 0
R1(config-if)#description R1 LAN
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#exit
R1(config)#interface Serial1/0
R1(config-if)#description R1 --> R2
R1(config-if)#ip address 10.1.1.1 255.255.255.252
R1(config-if)#clockrate 128000
R1(config-if)#no shutdown
R1(config-if)#exit
```

R2

```
R2#conf t
Enter configuration commands, one per line.  End with CNT
R2(config)#interface Serial1/0
R2(config-if)#description R2 --> R1
R2(config-if)#ip address 10.1.1.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface Serial1/1
R2(config-if)#description R2 --> R3
R2(config-if)#ip address 10.2.2.1 255.255.255.252
R2(config-if)#clockrate 128000
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
```

R3

```
R3#conf t
Enter configuration commands, one per line.  End with CNT
R3(config)#interface Loopback0
R3(config-if)#description R3 LAN
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#exit
R3(config)#interface Serial1/1
R3(config-if)#description R3 --> R2
R3(config-if)#ip address 10.2.2.2 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
```

Step 2: Configure static routes.

R1

```
R1(config)#
R1(config)#ip route 0.0.0.0 0.0.0.0 10.1.1.2
R1(config)#exit
R1#
```

R3

```
R3(config)#
R3(config)#ip route 0.0.0.0 0.0.0.0 10.2.2.1
R3(config)#
```

R2

```
*Apr 19 13:53:28.787: %LINEPROTO-5-UPDOWN: Line protoco
 changed state to up
R2(config)#
R2(config)#ip route 192.168.1.0 255.255.255.0 10.1.1.1
R2(config)#ip route 192.168.3.0 255.255.255.0 10.2.2.2
R2(config)#
```

From the R1 router, run the following Tcl script to verify connectivity.

```
R1
R1#tclsh
R1(tcl)#foreach address {
+>(tcl)#192.168.1.1
+>(tcl)#10.1.1.1
+>(tcl)#10.1.1.2
+>(tcl)#10.2.2.1
+>(tcl)#10.2.2.2
+>(tcl)#192.168.3.1
+>(tcl)#} { ping $address }

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/4 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
....!
Success rate is 20 percent (1/5), round-trip min/avg/max = 1004/1004/1004 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/31/32 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/32/40 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/62/72 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/63/64 ms
R1(tcl)#
```

Step 3: Secure management access.

```
R1
R1(tcl)#exit
R1#conf t
Enter configuration commands, one per line.  End with CNT
R1(config)#security passwords min-length 10
R1(config)#enable secret class12345
R1(config)#line console 0
R1(config-line)#password mscit25260
R1(config-line)#exec-timeout 5 0
R1(config-line)#login
R1(config-line)#logging sync
R1(config-line)#exit
R1(config)#line vty 0 4
R1(config-line)#password mscit25261
R1(config-line)#exec-timeout 5 0
R1(config-line)#login
R1(config-line)#exit
R1(config)#line aux 0
R1(config-line)#no exec
R1(config-line)#end
```

```
R1
R1(config)#service password-encryption
```

Step 4: Configure enhanced username password security in R1 and R3.

```
R1
R1(config)#banner motd $Unauthorized access strictly prohibited!$
R1(config)#username JR-ADMIN secret class12345
R1(config)#username ADMIN secret class54321
R1(config)#line console 0
R1(config-line)#login local
R1(config-line)#exit
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#exit
R1(config)#exit
```

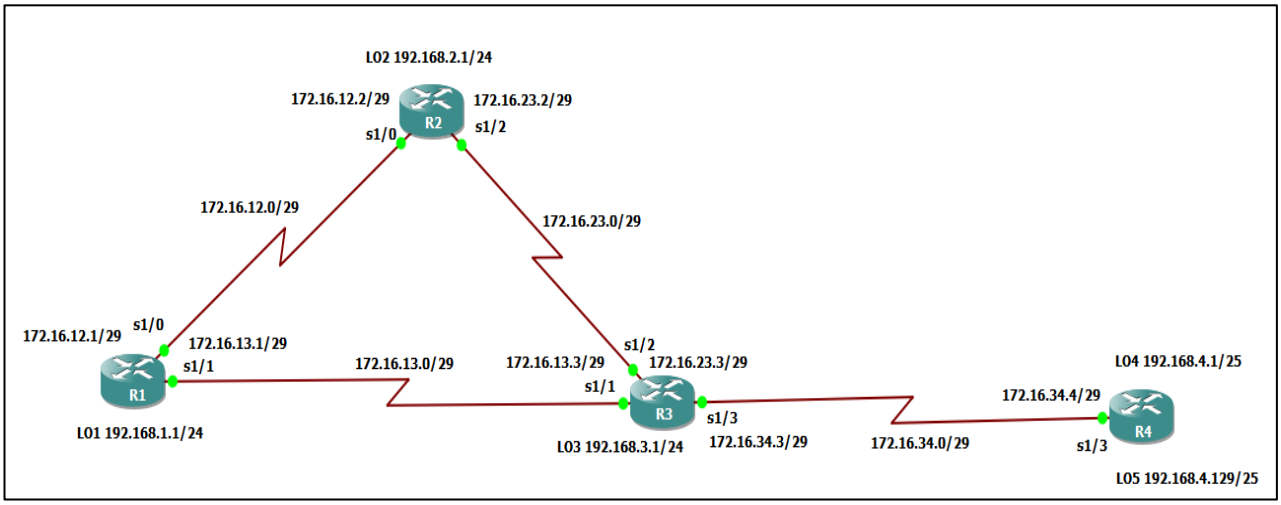
```
R3
R3(config)#
R3(config)#username JR-ADMIN secret class12345
R3(config)#username ADMIN secret class54321
R3(config)#line console 0
R3(config-line)#login local
R3(config-line)#exit
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#exit
R3(config)#exit
```

Step 5: Try to access router 3 from router 1.

```
R1#telnet 10.2.2.2
Trying 10.2.2.2 ... Open


User Access Verification

Username: ADMIN
Password:
R3>
```

Practical No 5:

Aim: Configure and Verify Path Control Using PBR Topology



Step 1: Configure loopbacks and assign addresses.

```
R1
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTI
R1(config)#
R1(config)#interface Lo1
R1(config-if)#description R1 LAN
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#exit
R1(config)#interface Serial1/0
R1(config-if)#description R1 --> R2
R1(config-if)#ip address 172.16.12.1 255.255.255.248
R1(config-if)#clockrate 128000
R1(config-if)#bandwidth 128
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface Serial1/1
R1(config-if)#description R1 --> R3
R1(config-if)#ip address 172.16.13.1 255.255.255.248
R1(config-if)#bandwidth 64
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
```

R2

```
 changed state to down
R2#
R2#conf t
Enter configuration commands, one per line.  End with CNTL
R2(config)#interface Lo2
R2(config-if)#description R2 LAN
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#interface Serial1/0
R2(config-if)#description R2 --> R1
R2(config-if)#ip address 172.16.12.2 255.255.255.248
R2(config-if)#bandwidth 128
R2(config-if)#no shutdown
R2(config-if)#interface Serial1/2
R2(config-if)#description R2 --> R3
R2(config-if)#ip address 172.16.23.2 255.255.255.248
R2(config-if)#clockrate 128000
R2(config-if)#bandwidth 128
R2(config-if)#no shutdown
R2(config-if)#end
R2#
```

R3

```
R3#conf t
Enter configuration commands, one per line.  End with CNTL/
R3(config)#interface Lo3
R3(config-if)#description R3 LAN
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#interface Serial1/1
R3(config-if)#description R3 --> R1
R3(config-if)#ip address 172.16.13.3 255.255.255.248
R3(config-if)#clock rate 64000
R3(config-if)#bandwidth 64
R3(config-if)#no shutdown
R3(config-if)#interface Serial1/2
R3(config-if)#description R3 --> R2
R3(config-if)#ip address 172.16.23.3 255.255.255.248
R3(config-if)#bandwidth 128
R3(config-if)#no shutdown
R3(config-if)#interface Serial1/3
R3(config-if)#description R3 --> R4
R3(config-if)#ip address 172.16.34.3 255.255.255.248
R3(config-if)#clock rate 64000
R3(config-if)#bandwidth 64
R3(config-if)#no shutdown
R3(config-if)#end
```

```
R4
R4#
R4#conf t
Enter configuration commands, one per line.  End with CNTL/Z
R4(config)#interface Lo4
R4(config-if)#description R4 LAN A
R4(config-if)#ip address 192.168.4.1 255.255.255.128
R4(config-if)#interface Lo5
R4(config-if)#description R4 LAN B
R4(config-if)#ip address 192.168.4.129 255.255.255.128
R4(config-if)#interface Serial1/3
R4(config-if)#description R4 --> R3
R4(config-if)#ip address 172.16.34.4 255.255.255.248
R4(config-if)#bandwidth 64
R4(config-if)#no shutdown
R4(config-if)#end
R4#
```

```
R3
*Apr 19 17:22:49.923: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip interface brief | include up
Serial1/1                  172.16.13.3     YES manual up                    up

Serial1/2                  172.16.23.3     YES manual up                    up

Serial1/3                  172.16.34.3     YES manual up                    up

Loopback3                  192.168.3.1     YES manual up                    up

R3#conf t
```

Step 3: Configure basic EIGRP.

```
R1
Enter configuration commands, one per line.  End with CNTI
R1(config)#router eigrp 1
R1(config-router)#network 192.168.1.0
R1(config-router)#network 172.16.12.0 0.0.0.7
R1(config-router)#network 172.16.13.0 0.0.0.7
R1(config-router)#no auto-summary
R1(config-router)#
```

```
R2
R2#conf t
Enter configuration commands, one per line.  End with CNT
R2(config)#router eigrp 1
R2(config-router)#network 192.168.2.0
R2(config-router)#network 172.16.12.0 0.0.0.7
R2(config-router)#network 172.16.23.0 0.0.0.7
R2(config-router)#no auto-summary
R2(config-router)#
```

```
R3
R3#conf t
Enter configuration commands, one per line.  End with CN
R3(config)#router eigrp 1
R3(config-router)#network 192.168.3.0
R3(config-router)#network 172.16.13.0 0.0.0.7
R3(config-router)#network 172.16.23.0 0.0.0.7
R3(config-router)#network 172.16.34.0 0.0.0.7
R3(config-router)#no auto-summary
R3(config-router)#
```

```
R4
R4#conf t
Enter configuration commands, one per line.  End with CNTL
R4(config)#router eigrp 1
R4(config-router)#network 192.168.4.0
R4(config-router)#network 172.16.34.0 0.0.0.7
R4(config-router)#no auto-summary
R4(config-router)#
```

Step 4: Verify the current path.

```
R4
R4(config-router)#exit
R4(config)#exit
R4#
*Apr 19 17:28:45.263: %SYS-5-CONFIG_I: Configured from con
R4#traceroute 192.168.1.1 source 192.168.4.1

Type escape sequence to abort.
Tracing the route to 192.168.1.1

  1  *  *
    172.16.34.3 1080 msec
  2 172.16.23.2 64 msec 64 msec 64 msec
  3 172.16.12.1 76 msec 80 msec 80 msec
R4#traceroute 192.168.1.1 source 192.168.4.129

Type escape sequence to abort.
Tracing the route to 192.168.1.1

  1 172.16.34.3 580 msec 28 msec 32 msec
  2 172.16.23.2 72 msec 68 msec 64 msec
  3 172.16.12.1 80 msec 76 msec 64 msec
```

Notice that the path taken for the packets sourced from the R4 LANs are going through R3 --> R2 -->R1.

Step 5: Configure PBR to provide path control.

```
R3
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip access-list standard PBR-ACL
R3(config-std-nacl)#remark ACL matches R4 LAN B traffic
R3(config-std-nacl)#permit 192.168.4.128 0.0.0.127
R3(config-std-nacl)#exit
R3(config)#
R3(config)#route-map R3-to-R1 permit
R3(config-route-map)#description RM to forward LAN B traffic to R1
R3(config-route-map)#match ip address PBR-ACL
R3(config-route-map)#set ip next-hop 172.16.13.1
R3(config-route-map)#exit
```

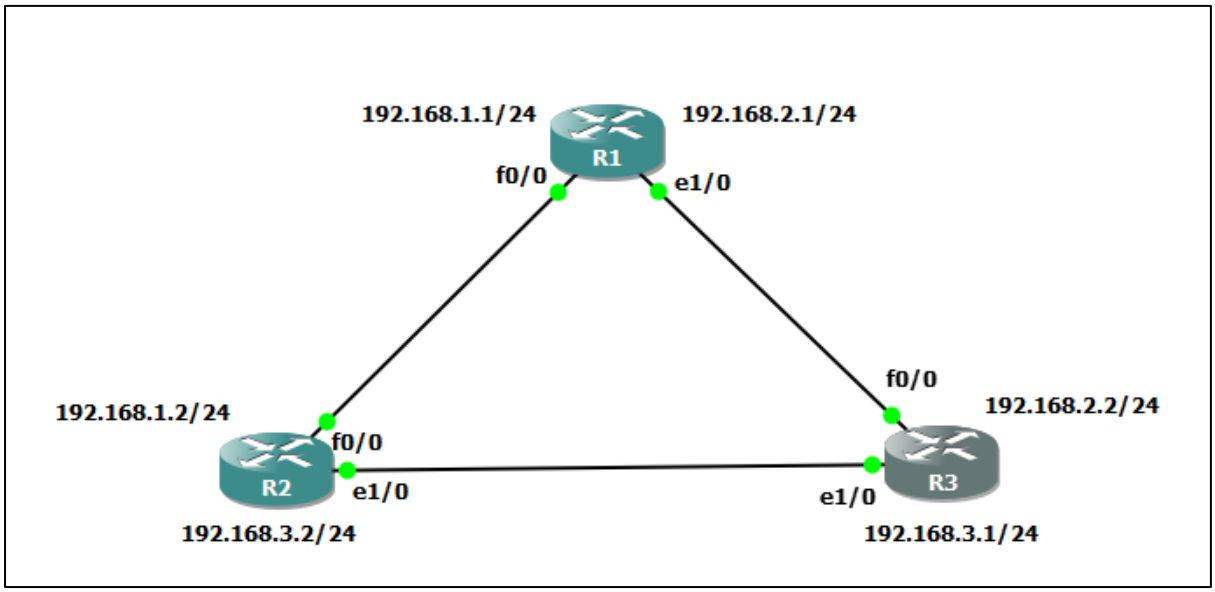
```
R3
R3#conf t
Enter configuration commands, one per line.
R3(config)#int s1/3
R3(config-if)#ip policy route-map R3-to-R1
R3(config-if)#end
R3#
```

Step 6: Test the policy.

```
R4
  3 172.16.12.1 80 msec 80 msec 72 msec
R4#traceroute 192.168.1.1 source 192.168.4.1

Type escape sequence to abort.
Tracing the route to 192.168.1.1

  1  *  *  *
  2 172.16.23.2 1000 msec 64 msec 64 msec
  3 172.16.12.1 76 msec 80 msec 80 msec
R4#traceroute 192.168.1.1 source 192.168.4.129

Type escape sequence to abort.
Tracing the route to 192.168.1.1

  1 172.16.34.3 500 msec 28 msec 32 msec
  2 172.16.13.1 68 msec 56 msec 68 msec
R4#
```

Practical No 6:

Aim: IP Service Level Agreements and Remote SPAN in a Campus Environment.



Step 1: Config all routers with their respective IP address and Loopback address according to the topology.

R1

```
R1#
R1#conf t
Enter configuration commands, one per line.  End with
R1(config)#int f0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shut
```

R1

```
*Apr 19 19:02:01.303: %LINEPROTO-5-UPDOWN: Line protocol c
et0/0, changed state to up
R1(config)#int e1/0
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shut
```

R2

```
R2#
R2#conf t
Enter configuration commands, one per line.  End with
R2(config)#int f0/0
R2(config-if)#ip address 192.168.1.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
```

```
R2
et0/0, changed state to up
R2(config)#int e1/0
R2(config-if)#ip address 192.168.3.2 255.255.255.0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
```

```
R3

R3(config)#int f0/0
R3(config-if)#ip address 192.168.2.2 255.255.255.0
R3(config-if)#no shut
R3(config-if)#e
*Apr 19 19:06:04.339: %LINK-3-UPDOWN: Interface FastEth
o up
R3(config-if)#exit
R3(config)#
*Apr 19 19:06:04.339: %ENTITY_ALARM-6-INFO: CLEAR INFO
istrative State Down
*Apr 19 19:06:05.419: %LINEPROTO-5-UPDOWN: Line protoco
et0/0, changed state to up
R3(config)#int e1/0
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
```

Step 2: To connect with R2 N/W

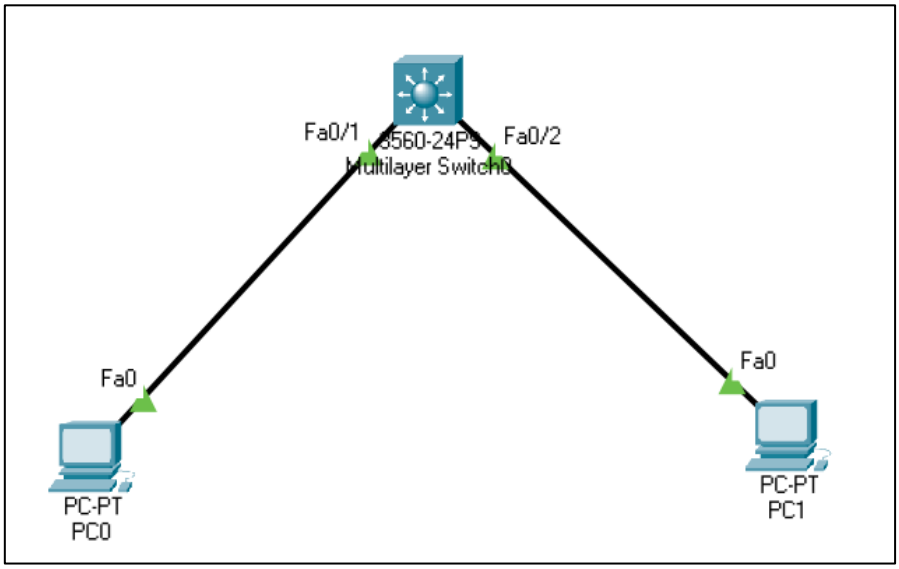
```
R2
0, changed state to up
R2(config)#ip route 192.168.2.0 255.255.255.0 192.168.1.1
R2(config)#ip route 192.168.2.0 255.255.255.0 192.168.3.1
R2(config)#track 10 ip route 192.168.2.0 255.255.255.0 reachability
R2(config-track)#exit
R2(config)#exit
R2#
```

Step 3: Verify route connection

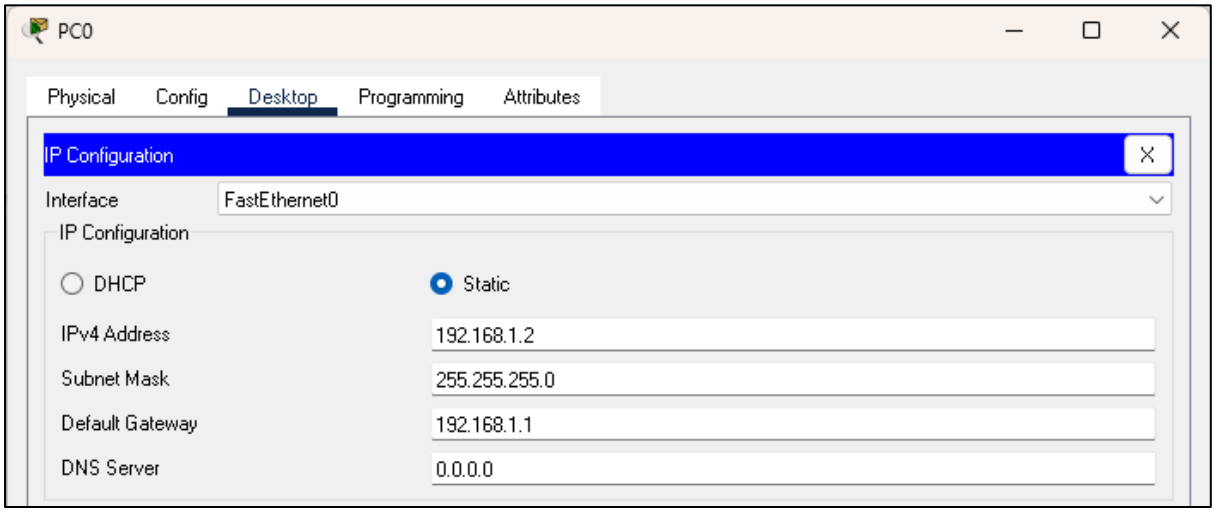
```
R2
R2#
*Apr 19 19:07:47.923: %SYS-5-CONFIG_I: Configured from console by console
R2#show track
Track 10
  IP route 192.168.2.0 255.255.255.0 reachability
  Reachability is Up (static)
    1 change, last change 00:00:12
  First-hop interface is FastEthernet0/0
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 192.168.3.1
                    [1/0] via 192.168.1.1
C    192.168.3.0/24 is directly connected, Ethernet1/0
R2#
```

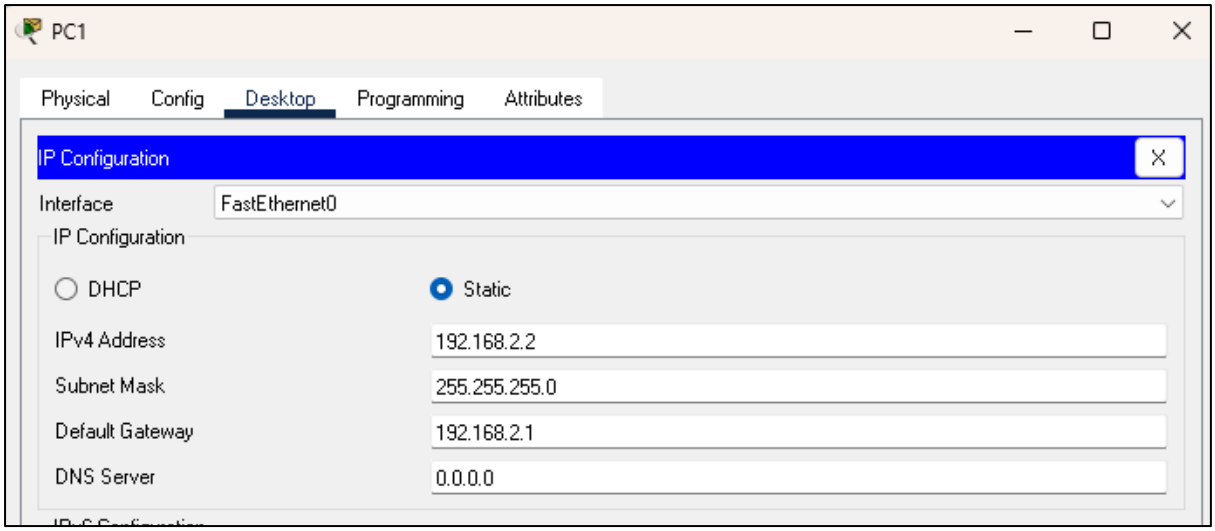
Practical No 7:

Aim: Inter-VLAN Routing



Step 1: Configuring IP address to PC0 and PC1





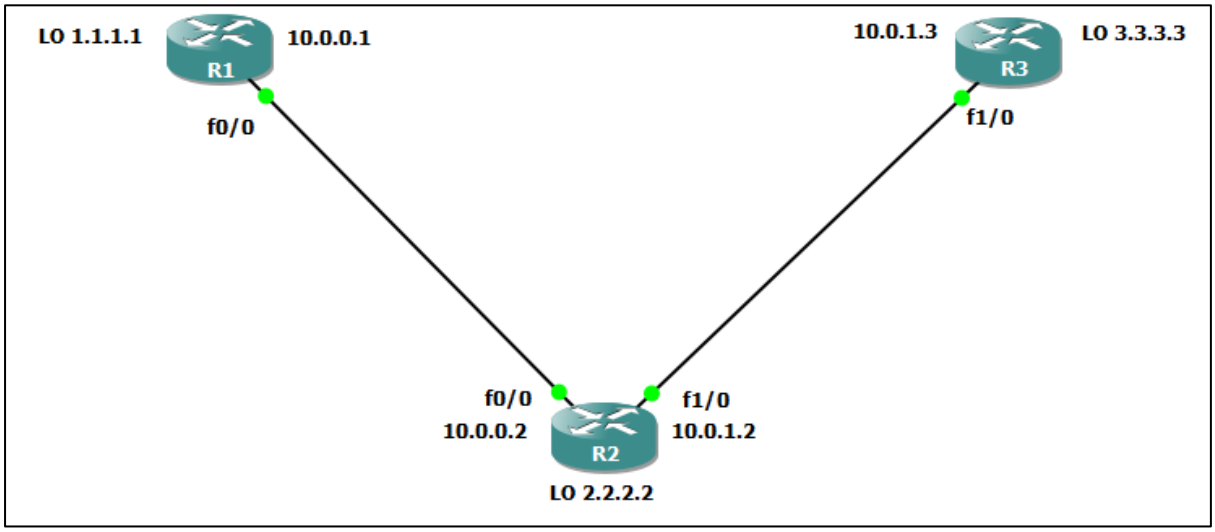

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name abc
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name abc
VLAN #10 and #20 have an identical name: abc
Switch(config-vlan)#exit
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#exit
```

```
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

Switch(config-if)#ip address 192.168.2.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#exit
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#
```

Practical No 8:

Aim: Simulating MPLS environment



Step 1: Configure loopbacks and assign addresses with ospf commands.

```
R1
R1(config)#int f0/0
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#ip ospf 1 area 0
R1(config-if)#
*Apr 19 19:46:54.859: %OSPF-4-NORTRID: OSPF process 1 failed to allocate unique
router-id and cannot start
R1(config-if)#no shut
R1(config-if)#
*Apr 19 19:46:59.783: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
R1(config-if)#
*Apr 19 19:46:59.783: %ENTITY_ALARM-6-INFO: CLEAR INFO Fa0/0 Physical Port Admin
istrative State Down
*Apr 19 19:47:00.783: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R1(config-if)#exit
R1(config)#int LO0
R1(config-if)#ip address
*Apr 19 19:49:38.279: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
 changed state to up
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#ip ospf 1 area 0
R1(config-if)#no shut
R1(config-if)#exit
```

R2

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#in LO0
R2(config-if)#ip adre
*Apr 19 19:50:14.715: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
 changed state to up
R2(config-if)#ip address 2.2.2.2 255.255.255.255
R2(config-if)#ip ospf 1 area 0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
R2(config)#int f0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.0
R2(config-if)#ip ospf 1 area 0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#
*Apr 19 19:51:50.375: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
R2(config)#
*Apr 19 19:51:50.375: %ENTITY_ALARM-6-INFO: CLEAR INFO Fa0/0 Physical Port Admin
istrative State Down
*Apr 19 19:51:51.375: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
R2(config)#
*Apr 19 19:51:57.951: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.0.1 on FastEthernet0/0
 from LOADING to FULL, Loading Done
R2(config)#int f1/0
R2(config-if)#ip address 10.0.1.2 255.255.255.0
R2(config-if)#ip ospf 1 area 0
R2(config-if)#no shut
R2(config-if)#
```

R3

```
R3#
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#int LO0
R3(config-if)#ip address
*Apr 19 19:53:59.335: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
 changed state to up
R3(config-if)#ip address 3.3.3.3 255.255.255.255
R3(config-if)#ip ospf 1 area 0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int f1/0
R3(config-if)#ip address 10.0.1.3 255.255.255.0
R3(config-if)#ip ospf 1 area 0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#
*Apr 19 19:55:12.123: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
R3(config)#
*Apr 19 19:55:12.123: %ENTITY_ALARM-6-INFO: CLEAR INFO Fa1/0 Physical Port Admin
istrative State Down
*Apr 19 19:55:13.123: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
```

Step 2: Check for ospf configuration and include the state up.

```
R1
R1(config)#
*Apr 19 19:52:07.943: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0
from LOADING to FULL, Loading Done
R1(config)#do sh ip ospf int br
Interface     PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Lo0           1     0               1.1.1.1/32         1     LOOP  0/0
Fa0/0         1     0               10.0.0.1/24        1     DR    1/1
R1(config)#do sh ip int br | include up
FastEthernet0/0            10.0.0.1        YES manual up                    up

Loopback0                  1.1.1.1         YES manual up                    up
```

Step 3: Verify the connection.

```
R3
*Apr 19 19:56:53.211: %SYS-5-CONFIG_I: Configured from console by console
R3#ping 1.1.1.1 source LO0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
Packet sent with a source address of 3.3.3.3
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/64/72 ms
R3#conf t
```

Step 4: Establish MPLS configuration.

```
R1
R1(config)#router ospf 1
R1(config-router)#mpls ldp autoconfig
R1(config-router)#
*Apr 19 19:58:45.299: %LDP-5-NBRCHG: LDP Neighbor 2.2.2.2:0 (1) is UP
R1(config-router)#
```

```
R2
R2(config)#router ospf 1
R2(config-router)#mpls autoconfig
                   ^
% Invalid input detected at '^' marker.

R2(config-router)#mpls ldp autoconfig
R2(config-router)#
```

```
R3
R3#conf t
Enter configuration commands, one per line.  End with
R3(config)#router ospf 1
R3(config-router)#mpls ldp autoconfig
R3(config-router)#
*Apr 19 19:58:17.623: %LDP-5-NBRCHG: LDP Neighbor 2.2.
```

Step 5: Verify Established connection of MPLS

```
R2
*Apr 19 19:58:51.211: %SYS-5-CONFIG_I: Configured from console by
R2#sh mpls interface
Interface              IP            Tunnel   Operational
FastEthernet0/0        Yes (ldp)     No       Yes
FastEthernet1/0        Yes (ldp)     No       Yes
R2#
```